Unit Test Report

Vehicle Parking System | Team 10 | Software Engineering Mini Project

Covers: unit-level tests of models and validation helpers (tests/test_models.py)

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Build tested	main branch, Team10-VPS/vehicle_parking_system (zip snapshot)
Environment	Python 3.12, Flask 3, Flask-SQLAlchemy, SQLite, pytest 9, coverage 7

1. Summary

Total tests (all pytest items)	99
Unit tests	20
Integration tests	79
Passed	99
Failed	0
Errors / skipped	0 / 0
Total run time	17.7 s

Command used: `SECRET_KEY=x pytest -v --junitxml=report.xml`. All tests passed, so no defects were raised.

2. Results by test file

File	Level	Tests	Passed	Failed	Time (s)
test_auth_integration.py	Integration	16	16	0	2.18
test_checkout_history_admin.py	Integration	26	26	0	6.40
test_csrf.py	Integration	2	2	0	0.02
test_models.py	Unit	20	20	0	0.69
test_slots_integration.py	Integration	24	24	0	5.39
test_vehicle_registration.py	Integration	11	11	0	2.77

3. Unit test cases (tests/test_models.py)

Test function	Cases	Test case ID	What it checks	Result
test_password_is_hashed_not_plain	1	TC-Auth-01, TC-Sec-02	Stored password is a hash, not plain text	Pass
test_check_password_correct_and_wrong	1	TC-Auth-02	Verification works only for the right password	Pass
test_same_password_gives_different_hashes	1	TC-Sec-02	Hashing is salted	Pass
test_duplicate_email_rejected_by_db	1	TC-Reg-02	Unique email constraint	Pass
test_default_role_is_customer	1	TC-Sec-03	New users default to customer role	Pass
test_slot_defaults_to_available	1	TC-Slot-01	New slot is available by default	Pass
test_duplicate_slot_number_rejected	1	TC-Admin-01	Unique slot number constraint	Pass
test_valid_email	6	TC-Sec-04	Valid and invalid email formats	Pass (6/6)
test_valid_plate	7	TC-Sec-04, TC-Book-02	Valid plates, empty, too short/long, script text, only symbols	Pass (7/7)

4. Observations

Unit coverage of the model layer is complete: app/models.py and app/utils.py are both at 100% statement coverage (see Code Coverage Report). The route level behaviour is tested by the 79 integration tests listed in the RTM sheet of Team10_TestCases_RTM.xlsx.